ROLL NO : 241801038

# **Exp 5**: Experiments on Packet Capture Tool – Wireshark

## AIM :

To capture, filter, view and analyse network packets using the Wireshark packet capture tool.

## ALGORITHM :

1. Start the experiment.
2. Install and launch Wireshark, and select the active network interface (Local Area Connection / Wi-Fi) under Capture.
3. Open Capture > Options and set the capture to stop automatically after 100 packets.
4. Click Start to begin capturing; the packets appear in real time in the Packet List pane.
5. Apply a display filter (tcp, udp, arp, dns, http, icmp, dhcp) in the filter bar to show only the required protocol.
6. Select a packet and inspect its headers in the Packet Details pane and its raw data in the Packet Bytes pane.
7. Use Statistics > Flow Graph to view the packet flow between the hosts.
8. Use View > Coloring Rules to understand the colour coding of the different protocols.
9. Save the captured packets using File > Save.
10. Stop the experiment.

## PROCEDURE :

Wireshark is a packet sniffer and network analysis tool, formerly known as Ethereal. It captures packets in real time and displays them in a human-readable format. It is a passive program: it never sends packets itself and receives a copy of all packets sent and received by the computer.

**Capturing 100 packets from the Ethernet (IEEE 802.3) LAN interface and saving them:**

- Select Local Area Connection in Wireshark.
- Go to Capture → Options.
- Select “Stop capture automatically after 100 packets”.
- Click Start capture.
- Save the packets (File → Save).

**Display filters (capture 100 packets as above, then type the filter in the filter bar and press Enter):**

- **TCP / UDP** – type tcp or udp in the filter bar, inspect the packets, and click Statistics → Flow Graph to view the flow graph.
- **ARP** – type arp in the filter bar and inspect the packets.
- **DNS** – type dns in the filter bar and click Statistics → Flow Graph to view the flow graph.
- **HTTP** – type http in the filter bar and inspect the packets.

- **IP / ICMP** – type icmp (or ip) in the filter bar and inspect the packets.
- **DHCP** – type dhcp in the filter bar and inspect the packets.

**Colour coding:**

- Click View → Coloring Rules to see what each colour means. By default, light purple is TCP traffic, light blue is UDP traffic, and black identifies packets with errors (for example, delivered out of order).
- Save the captured packets using File → Save.

# OUTPUT :

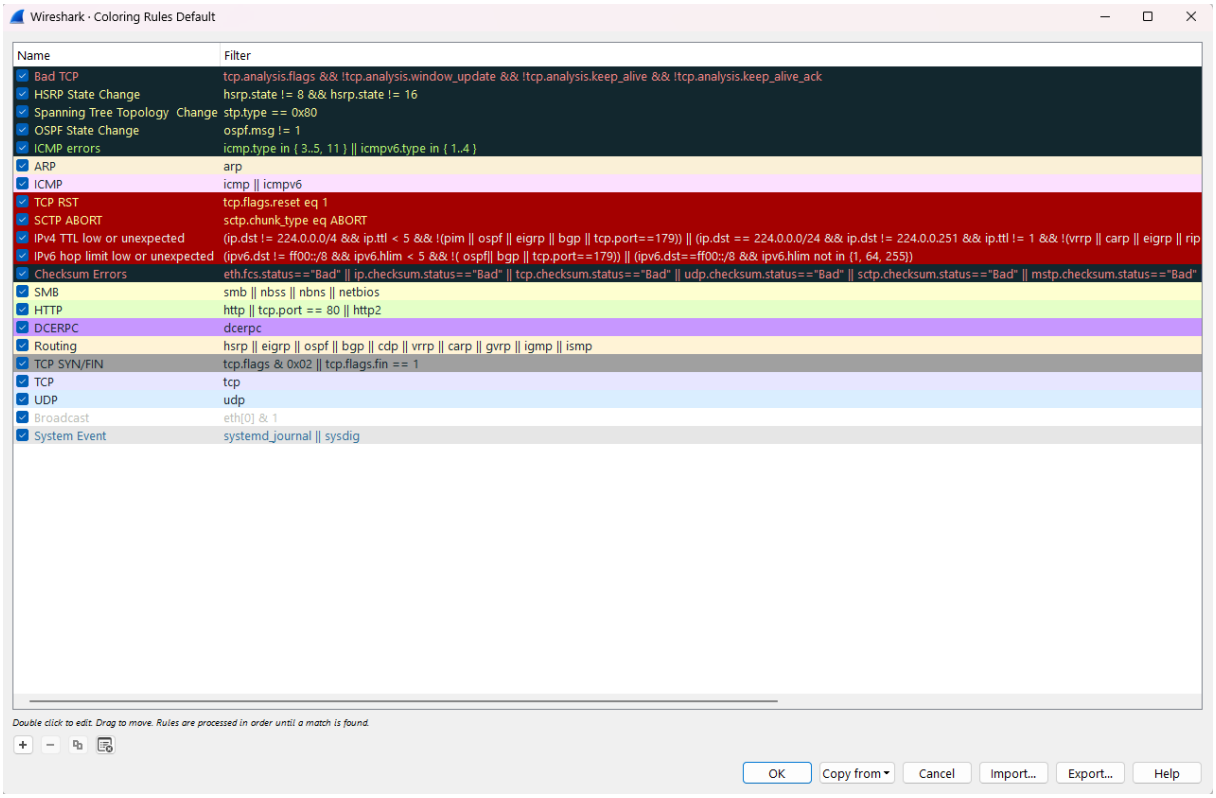



*Fig 1. Wireshark – Coloring Rules (View → Coloring Rules)*

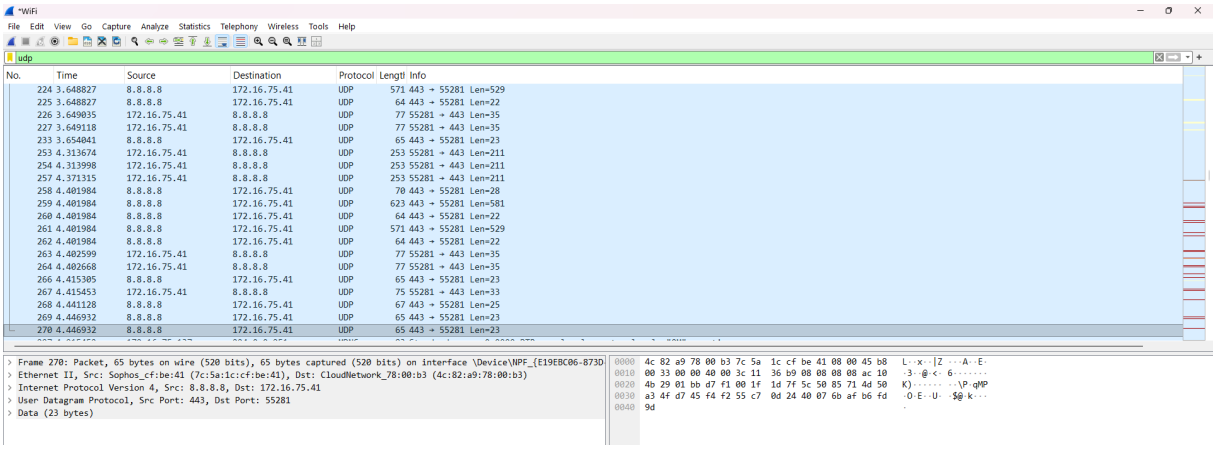



*Fig 2. UDP packets captured using the display filter “udp”*

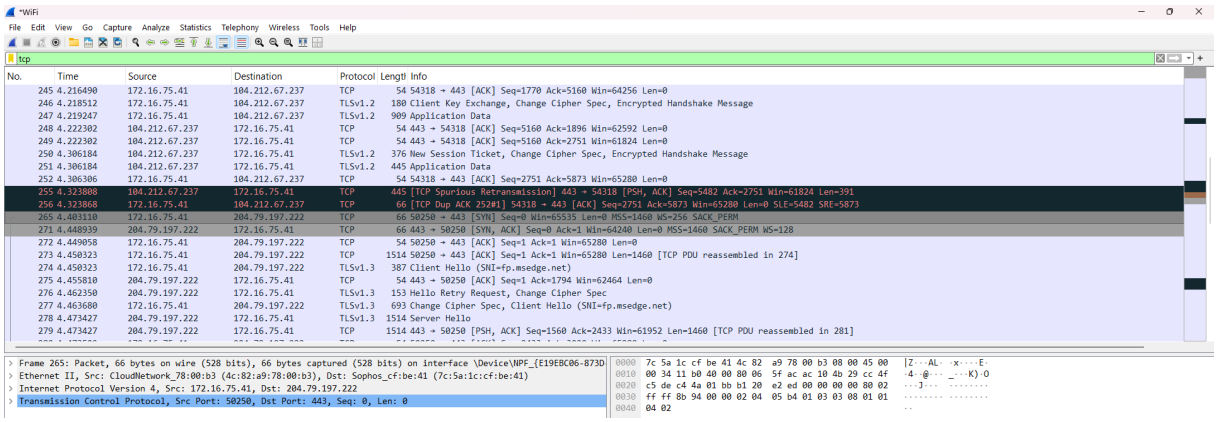



*Fig 3. TCP packets captured using the display filter “tcp”*

# RESULT :

Thus, packets were successfully captured and analysed using the Wireshark packet capture tool. TCP and UDP packets were filtered and inspected, and the colour coding rules of Wireshark were studied.